Christophe de Dinechin

Software as a Passion

French software developer — operating systems, languages, games

SYSTEMS & VIRTUALIZATION

HPVM - Enterprise-grade Itanium virtualization 2000-2010

- From first line of code and pre-silicon bring up all the way to 7 digits of revenue per quarter

HP ECUTEST Real Time Test System for car electronics 1994-1998

- Engine control unit (ECU) real-time testing and validation, 400 analog and digital I/O signals

SPICE Remote Streaming 2017-Present

- Bring the SPICE remote-access application to the modern 3D and video desktop

DxO ONE Connected Camera 2015-2016

- A tiny, but high-quality camera for photographers. Dual-core ARM + ISP, RTOS + Linux

LANGUAGES & COMPILERS

HP aC++ Compiler 1998-2000

- First native pre-silicon Itanium compiler, exception handling, templates, performance, build system

C++ ABI 1999-2000

- Initiated a cross-industry standardization effort for C++ compiler interoperability, still in use today

XL 1992-Present

- XL is an Extensible Language with interpreted and JIT modes

XLR8

- XLR8 is a super-fast VM, with JIT to ARM, x86 and RISC-V, 32 and 64 bits and full LLVM support

DB48X - Modern RPL Calculator 2022-Present

- Best scientific calculator today, a modern rebuild of high-end HP-48 and HP-50 calculators

TOOLS & APPS

Make-It-Quick

- A make-based build system with auto-configuration, replaces automake/autoconf/cmake

Recorder

- A flight recorder for C and C++ programs, printf syntax, continuous logging, real-time graphing

Emacs on Aqua

- First graphical port of Emacs on macOS X (Rhapsody), first Emacs with transparent windows

GAMES & 3D

Alpha Waves / Continuum 1989-1990

- First 3D platform game ever (Guinness Book of Records), precursor of Alone in the Dark

HP-48 Games 1989-1993

- Pacman, Tetris, Lemmings, demos for a cross-compiler I had written, went viral over infrared

Tao3D 2010-Present

- Tao3D is an XL-based platform for interactive real-time 3D presentations

Taodyne 2010-2016

- A company selling Tao3D and high-end interactive 3D presentations, stereoscopy, glasses-free 3D

RESEARCH

Computer science research

- Published CS research: WIESS 2000 C++ exception handling; FOSDEM 2024 / FOSDEM 2023 talks on DB48X

12 patents awarded, 3 pending

- 12 granted US patents, 3 pending

Physics research

- A Theory of Incomplete Measurements — unifying GR and QM (TIM, Réunifions la physique)